Experiment 10 — Import and Explore a Dataset

Python Lab Manual — Program and Output

EX. NO: 10(a) — Explore a dataset using Pandas

PROGRAM

```
import numpy as np
import pandas as pd

data = {
    "animal": ["cat", "cat", "snake", "dog", "dog", "cat", "snake", "cat", "dog", "dog"],
    "age": [2.5, 3, 0.5, np.nan, 5, 2, 4.5, np.nan, 7, 3],
    "visits": [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    "priority": ["yes", "yes", "no", "yes", "no", "no", "no", "yes", "no", "no"]
}
labels = ["a", "b", "c", "d", "e", "f", "g", "h", "i", "j"]
df = pd.DataFrame(data, index=labels)

print(df)
df.info()
print(df.head(3))
print(df.loc[df.index[[3, 4, 8]], ["animal", "age"]])
print(df["visits"] > 2)
print(df[df["age"].isnull()])
print(df[(df["animal"] == "cat") & (df["age"] < 3)])
print(df[df["age"].between(2, 4)])
print(df.groupby("animal")["age"].mean())
print(df["animal"].value_counts())
print(df.sort_values(by=["age", "visits"], ascending=[False, True]))
```

OUTPUT / RESULT

```
Available outputs shown in the supplied PDF include:
visits > 2:
a    False
b     True
c    False
d     True
e    False
f     True
g    False
h    False

Mean age by animal:
cat      2.5
dog      5.0
snake    2.5

Several other output areas are blank in the supplied PDF.
```

EX. NO: 10(b) — Explore a dataset using NumPy

PROGRAM

```
import numpy as np

L = list(range(10))
print([str(c) for c in L])
print([type(item) for item in L])

print(np.zeros(10, dtype="int"))
print(np.ones((3, 5), dtype=float))
print(np.full((3, 5), 1.23))
print(np.arange(0, 20, 2))
print(np.linspace(0, 1, 5))
print(np.random.normal(0, 1, (3, 3)))
print(np.eye(3))

np.random.seed(0)
x1 = np.random.randint(10, size=6)       # one dimension
x2 = np.random.randint(10, size=(3, 4))  # two dimensions
x3 = np.random.randint(10, size=(3, 4, 5))  # three dimensions

print("x3 ndim:", x3.ndim)
print("x3 shape:", x3.shape)
```

```
print("x3 size:", x3.size)
```

OUTPUT / RESULT

```
['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']
[int, int, int, int, int, int, int, int, int, int]
array([0, 0, 0, 0, 0, 0, 0, 0, 0])
x3 ndim: 3
x3 shape: (3, 4, 5)
x3 size: 60
```